Mississippi Valley State University

# CS 341 — Discrete Structures

## Lecture 38: Graphs as Data

*Adjacency Matrices and Families of Graphs*

Chartrand & Zhang, Section 12.1

Name: _____ Date: _____

**Where we are**

A picture of a graph is convenient for people and useless to a computer. Today: the standard way to store a graph as an array of numbers — and a surprising fact about what happens when you multiply that array by itself.

## By the end of today you should be able to

1. Write the adjacency matrix of a graph and read a graph off a matrix.
2. Use $A^k$ to count walks.
3. Recognise $K_n$, $C_n$, $P_n$, and bipartite graphs.

## 1. The adjacency matrix

**Adjacency matrix**

For a graph $G$ of order $n$ with $V(G) = \{v_1, \dots, v_n\}$, the **adjacency matrix** is the $n \times n$ matrix $A = [a_{ij}]$ with

$$a_{ij} = \begin{cases} 1 & \text{if } v_i v_j \in E(G) \\ 0 & \text{otherwise.} \end{cases}$$

**Three properties, each with a reason.**

| Property | Why |
|---|---|
| The main diagonal is all 0 | No vertex is adjacent to itself. |
| $A$ is symmetric | $v_iv_j$ is the same edge as $v_jv_i$. |
| Row $i$ sums to $\deg v_i$ | It counts the neighbours of $v_i$. |

**Build one.** $V = \{v_1, \dots, v_4\}$, $E = \{v_1v_2, v_1v_3, v_2v_3, v_3v_4\}$.

| | $v_1$ | $v_2$ | $v_3$ | $v_4$ |
|---|---|---|---|---|
| $v_1$ | 0 | 1 | 1 | 0 |
| $v_2$ | 1 | 0 | 1 | 0 |
| $v_3$ | 1 | 1 | 0 | 1 |
| $v_4$ | 0 | 0 | 1 | 0 |

Total of all entries: $8 = 2m$, so the size is $4$.

## 2. Powers of $A$ count walks

**Walk**

A **walk** of length $k$ from $u$ to $v$ is a sequence of $k+1$ vertices starting at $u$ and ending at $v$, with consecutive vertices adjacent. Vertices and edges *may* repeat.

**Theorem 12.22**

For a graph with adjacency matrix $A$, the number of $v_i$–$v_j$ walks of length $k$ is the $(i,j)$ entry of $A^k$.

For the matrix above,

$$A^2 = \begin{bmatrix} 2 & 1 & 1 & 1 \\ 1 & 2 & 1 & 1 \\ 1 & 1 & 3 & 0 \\ 1 & 1 & 0 & 1 \end{bmatrix} \qquad A^3 = \begin{bmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 4 & 1 \\ 4 & 4 & 2 & 3 \\ 1 & 1 & 3 & 0 \end{bmatrix}$$

| Question | Answer |
|---|---|
| Walks of length 2 from $v_1$ to $v_4$ | 1 — namely $(v_1, v_3, v_4)$ |
| Why is $a_{33}^{(2)} = 3$? | It is $\deg v_3$: out and back along each of 3 edges. |
| Walks of length 3 from $v_2$ to $v_3$ | 4 |

**The diagonal of $A^2$ is the degree sequence**

A walk of length 2 from $v$ back to $v$ goes out along an edge and straight back, so there is one for each edge at $v$. That is why $a_{ii}^{(2)} = \deg v_i$ — and it gives a second way to read degrees off the matrix.

## 3. Families worth naming

| Name | Description | Order and size |
|---|---|---|
| $K_n$ | complete: every pair adjacent | order $n$, size $\binom{n}{2}$ |
| $C_n$ | cycle on $n$ vertices | order $n$, size $n$ |
| $P_n$ | path on $n$ vertices | order $n$, size $n-1$ |
| $K_{s,t}$ | complete bipartite | order $s+t$, size $st$ |

$K_n$ is $(n-1)$-regular, and its size $\binom{n}{2}$ is the handshake count from Week 13 — the same number, because an edge *is* an unordered pair.

**Bipartite**

$G$ is **bipartite** if $V(G)$ can be split into two sets so that every edge joins a vertex in one to a vertex in the other — no edge inside either set.

**A theorem worth knowing:** a nontrivial graph is bipartite **if and only if** it contains no odd cycle.

So $C_4$ is bipartite and $C_5$ is not. Checking for an odd cycle is far easier than searching for a split.

## Exit ticket

1. What is the size of $K_7$? Of $K_{3,4}$?

2. If $A$ is an adjacency matrix, what does the $(2,2)$ entry of $A^2$ tell you?

## Before Friday

**Read** Section 12.2, "Connected Graphs," pp. 444–455.

**Self-check (not collected).** Section 12.1, Exercises 9, 11, and 13 — odd-numbered, answers in the back.

**Homework 13 is due Friday, November 20, 11:59 p.m.**

## Questions I still have